

The empirical recurrence for OEIS A298316, recovered from its tail

Adrian Perez Fontelles

Independent researcher

8 September 2026

Abstract

OEIS A298316 records “Empirical recurrence of order 71” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 72$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 73$. Its last coefficient is $c_{71} = 128 \neq 0$, so no further tail satisfies anything shorter; any order-71 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A298316 is “Number of $n \times 4$ 0..1 arrays with every element equal to 0, 1, 3, 4 or 6 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

5, 13, 9, 63, 119, 243, 1086, 2782, ...

The entry states:

Empirical recurrence of order 71 (see link above)

As of the “Last modified” line on the live entry (Jan 17 2018 07:14:47, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 72$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 71: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 71.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 144$ exact terms from $n = 2$ on, it returns order 71 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{71} c_i a(n-i),$$

c_1	3	c_{19}	29207080	c_{37}	1388270995	c_{55}	−4778582
c_2	5	c_{20}	12155910	c_{38}	775336183	c_{56}	−1055021
c_3	50	c_{21}	−29364366	c_{39}	−538488505	c_{57}	667784
c_4	−197	c_{22}	−104679749	c_{40}	−1075055700	c_{58}	311577
c_5	−259	c_{23}	−38955824	c_{41}	−542155804	c_{59}	−247104
c_6	−946	c_{24}	104151934	c_{42}	328566082	c_{60}	−958550
c_7	5469	c_{25}	282710572	c_{43}	575892726	c_{61}	−391575
c_8	5653	c_{26}	107381078	c_{44}	252951595	c_{62}	59298
c_9	7784	c_{27}	−255093430	c_{45}	−125382567	c_{63}	193558
c_{10}	−86129	c_{28}	−592694948	c_{46}	−193079186	c_{64}	176660
c_{11}	−69909	c_{29}	−258296035	c_{47}	−65890171	c_{65}	75672
c_{12}	−3648	c_{30}	450363044	c_{48}	19392045	c_{66}	24792
c_{13}	867940	c_{31}	990147294	c_{49}	22649180	c_{67}	−400
c_{14}	553667	c_{32}	508060500	c_{50}	−5890454	c_{68}	−912
c_{15}	−519858	c_{33}	−601857144	c_{51}	5141031	c_{69}	−192
c_{16}	−5971185	c_{34}	−1326617443	c_{52}	14531976	c_{70}	−192
c_{17}	−3022992	c_{35}	−749360223	c_{53}	7277894	c_{71}	128
c_{18}	5352867	c_{36}	639402919	c_{54}	−1876680		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 73$, and it is the recurrence OEIS A298316 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{71} - c_1 x^{71-1} - \dots - c_{71}$. Its constant term is $-c_{71} = -128$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the

minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 71 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 71, so $\deg q = 71$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 71, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 26 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 71, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 128.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-71 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A298316.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.